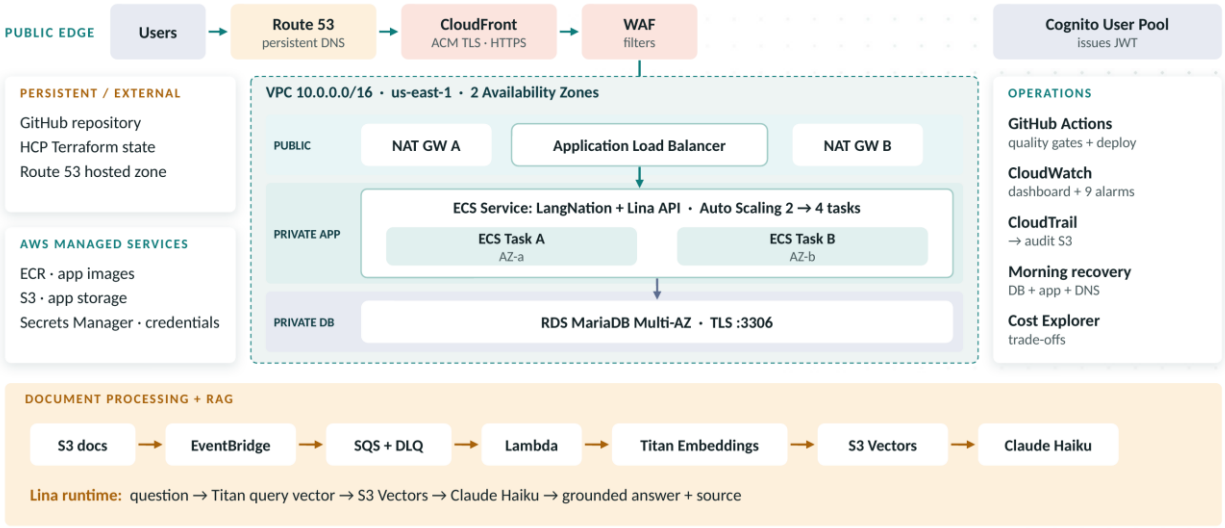
SIMPLE GUIDE TO EVERY MAIN TOOL & AWS SERVICE

LangNation Cloud Platform

What each tool is • why we used it • an easy way to remember it • where it appears in the project

Author	Seraj Ahmad Siddiqui
Course	AWS Cloud Computing with AI Bootcamp
Scope	Milestones 1–12 + automated sandbox recovery
AWS Region	us-east-1
Project endpoint	langnation.com/de
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How to use this guide

- Start with the big picture and the quick-reference table, then read the tool cards by category.
- Every card answers the same four questions: what it is, why we used it, an easy way to remember it, and where it appears in LangNation.
- Finish with the design decisions, final numbers and flows — these are the questions most likely to come up in a presentation.

Card colours

Development, IaC & automation	Networking	Containers & scaling	Data & storage
Events & RAG	Edge, security & auth	Monitoring & governance	

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1. First Understand the Big Picture

> **One-sentence architecture:** A user reaches LangNation through Route 53 → CloudFront/ACM/WAF → ALB → private ECS Fargate. The application uses private MariaDB and S3, Cognito for authentication, Bedrock + S3 Vectors for Lina RAG, CloudWatch and CloudTrail for operations/governance, Terraform/HCP Terraform for infrastructure, and GitHub Actions for quality-gated deployment and morning recovery.

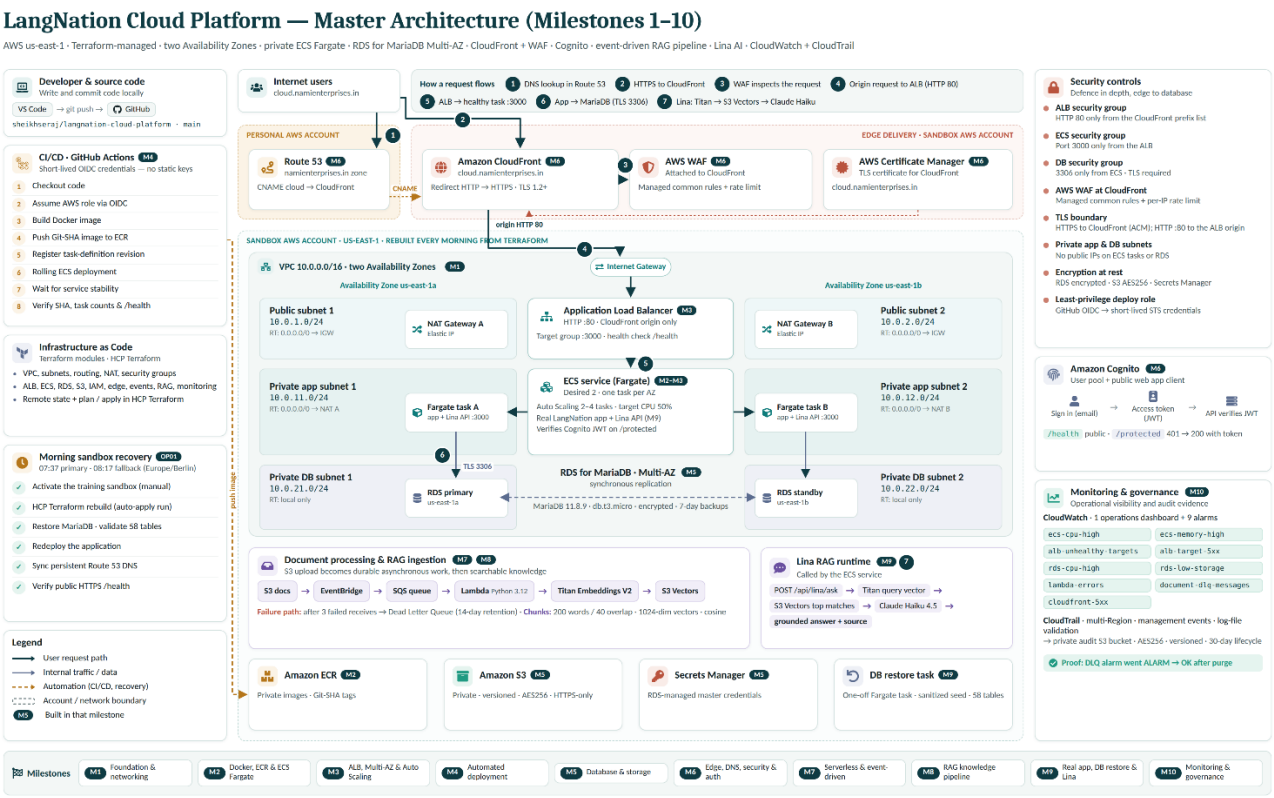


Figure 1. *Final LangNation Cloud Platform architecture (Milestones 1–12).*

Very simple user request flow

```
User → Route 53 → CloudFront (HTTPS + ACM) → AWS WAF → ALB → Target Group
  → private ECS Fargate → real LangNation application
    ├→ MariaDB
    ├→ S3
    ├→ Cognito
    └→ Lina RAG
```

Very simple RAG flow

```
Knowledge: S3 → EventBridge → SQS → Lambda → Titan embeddings → S3 Vectors
Question: ECS app → Titan query embedding → S3 Vectors → Claude Haiku
  → grounded answer
```

2. Quick-Reference Table

Area	Tool / Service	Full form / product	Main job
Development	VS Code	Visual Studio Code	Write and edit project code
Source control	Git / GitHub	Git + GitHub	Track code and store repository
Automation	GitHub Actions	CI/CD workflow platform	Quality gates, deploy, verify, recover
Containers	Docker	Container platform	Package app and restore task
Infrastructure	Terraform	Infrastructure as Code	Define AWS resources in code
Terraform cloud	HCP Terraform	HashiCorp Cloud Platform Terraform	Authoritative remote state/plan/apply
Command line	AWS CLI	AWS Command Line Interface	Inspect and validate AWS
Identity	IAM	Identity and Access Management	Permissions and roles
Federation	OIDC	OpenID Connect	Temporary GitHub-to-AWS identity
Networking	VPC	Virtual Private Cloud	Private AWS network
Networking	AZ	Availability Zone	Separate failure domain
Networking	IGW	Internet Gateway	Public internet path
Networking	NAT Gateway	Network Address Translation Gateway	Private outbound access
Networking	Elastic IP	Elastic IP address	Static public IPv4 address for each NAT Gateway
Security	Security Group	Security Group	Stateful network firewall
Registry	ECR	Elastic Container Registry	Store Docker images
Containers	ECS	Elastic Container Service	Orchestrate containers
Containers	Fargate	AWS Fargate	Serverless compute for ECS
Traffic	ALB	Application Load Balancer	Distribute HTTP requests
Scaling	Auto Scaling	Application Auto Scaling	Scale ECS tasks
Database	RDS for MariaDB	Relational Database Service	Managed private MariaDB
Secrets	Secrets Manager	AWS Secrets Manager	Store/manage DB credentials
Storage	S3	Simple Storage Service	Private object storage
Events	EventBridge	Amazon EventBridge	Route S3 Object Created events
Messaging	SQS	Simple Queue Service	Durable event buffer
Failure handling	DLQ	Dead Letter Queue	Preserve repeated failures
Serverless	Lambda	AWS Lambda	Process document events
AI	Bedrock	Amazon Bedrock	Invoke embedding and chat models
Embeddings	Titan Text Embeddings V2	Amazon Titan	Convert text to vectors
Vector search	S3 Vectors	Amazon S3 Vectors	Store/query semantic vectors
DNS	Route 53	Amazon Route 53	Stable public DNS

Area	Tool / Service	Full form / product	Main job
TLS	ACM	AWS Certificate Manager	HTTPS certificate
Edge	CloudFront	Amazon CloudFront	Global HTTPS front door
Web security	WAF	Web Application Firewall	Block malicious requests
Authentication	Cognito	Amazon Cognito	Sign-in and token issuance
Monitoring	CloudWatch	Amazon CloudWatch	Logs, metrics, dashboard, alarms
Audit	CloudTrail	AWS CloudTrail	Record AWS management actions
Container security	Trivy	Trivy	Scan image vulnerabilities
Cost	Cost Explorer	AWS Cost Explorer	Analyze AWS spend and usage

3. Development, IaC & Automation

VS Code

FULL FORM
Visual Studio Code

What it is A code editor used to write Terraform, JavaScript, Docker and workflow files.

Why we used it It gave us one place to build and manage the LangNation project locally.

Easy way to remember Like the desk where the engineer writes and organizes all project work.

In our project Used for Terraform modules, Node.js application code and GitHub Actions YAML.

Git

FULL FORM
Git

What it is A version-control system that records changes to files over time.

Why we used it It lets us track every infrastructure and application change and safely commit only intended files.

Easy way to remember Like a detailed history book with a restore point for every important change.

In our project Used for milestone commits, fixes and final documentation cleanup.

GitHub

FULL FORM
GitHub

What it is A hosted platform for Git repositories, collaboration and automation.

Why we used it It is the central source repository for LangNation and the trigger point for deployments.

Easy way to remember Like the shared central library that holds the official project copy.

In our project Repository main branch triggers selected workflows.

GitHub Actions

FULL FORM
GitHub Actions

What it is GitHub's workflow engine. It runs automated jobs when code is pushed or on a schedule.

Why we used it We use it to build Docker images, push them to ECR, deploy ECS, verify health and run morning recovery.

Easy way to remember Like a robot engineer that follows the same deployment checklist every time.

In our project Runs quality-gates.yml, deploy.yml and morning recovery.

Docker

FULL FORM
Docker

What it is Packages the application and its dependencies into a portable container image.

Why we used it It makes the Node.js application run consistently in development and on ECS Fargate.

Easy way to remember Like putting the app, runtime and instructions into one sealed shipping box.

In our project Used for the LangNation app and dedicated DB restore image.

Terraform

FULL FORM
Terraform

What it is A tool that describes cloud infrastructure using code instead of manual console clicks.

Why we used it Our sandbox is frequently cleaned, so Terraform lets us reproduce the whole environment reliably.

Easy way to remember Like an architectural blueprint that can rebuild the same building again and again.

In our project Modules manage networking, ECS, data, edge, events, RAG, monitoring and governance.

HCP Terraform

FULL FORM
HashiCorp Cloud Platform Terraform

What it is A managed remote Terraform service for plans, applies and state storage.

Why we used it It is our authoritative Terraform execution environment and keeps remote state safe.

Easy way to remember Like the project control room that stores the blueprint history and approves construction.

In our project Workspace cloudforge-capstone / langnation-capstone-dev.

AWS CLI

FULL FORM
Amazon Web Services Command Line Interface

What it is A command-line tool for querying and controlling AWS services.

Why we used it We used it to verify actual AWS state instead of trusting only the console.

Easy way to remember Like a direct command terminal for speaking to AWS.

In our project Used for ECS, ALB, SQS, Bedrock, CloudTrail and other validation.

IAM

FULL FORM
Identity and Access Management

What it is AWS service for users, roles and permissions.

Why we used it It controls exactly what GitHub Actions and AWS services are allowed to do.

Easy way to remember Like an access-control system that decides who has which keys.

In our project We created least-privilege deployment roles and an ECS task execution role.

OIDC FULL FORM OpenID Connect	**What it is** A standard that lets GitHub prove its identity to AWS and receive temporary credentials. **Why we used it** It avoids storing long-lived AWS access keys in GitHub. **Easy way to remember** Like showing a trusted temporary digital ID instead of carrying a permanent master key. **In our project** GitHub assumes one sandbox deployment role and one restricted personal-account Route 53 role.

4. Networking & Network Security

AWS Region

FULL FORM
Amazon Web Services Region

What it is A geographic AWS area containing multiple Availability Zones.

Why we used it We deploy the sandbox resources in us-east-1.

Easy way to remember Like choosing the city where your cloud infrastructure lives.

In our project Main sandbox region: us-east-1.

Availability Zone

FULL FORM
Availability Zone

What it is An isolated data-center location inside an AWS Region.

Why we used it Using two AZs makes the platform more available if one location has a problem.

Easy way to remember Like operating from two separate buildings in the same city.

In our project Uses us-east-1a and us-east-1b.

VPC

FULL FORM
Virtual Private Cloud

What it is A logically isolated private network inside AWS.

Why we used it It gives LangNation its own controlled network for public, application and database resources.

Easy way to remember Like a private campus with its own roads, entrances and security rules.

In our project CIDR 10.0.0.0/16.

Subnet

FULL FORM
Subnet

What it is A smaller network segment inside a VPC.

Why we used it We separate public resources, private application tasks and private databases for security.

Easy way to remember Like dividing a campus into public reception, staff offices and a locked records room.

In our project Six /24 subnets across two AZs.

Internet Gateway

FULL FORM
Internet Gateway

What it is Connects a VPC's public subnets to the Internet.

Why we used it The public ALB/NAT layer needs a controlled path to and from the Internet.

Easy way to remember Like the main front gate between the private campus and the public road.

In our project Public route table sends 0.0.0.0/0 to IGW.

Route Table

FULL FORM
Route Table

What it is Contains network routes that decide where packets should go.

Why we used it It gives public and private subnets different network behavior.

Easy way to remember Like road signs that tell traffic which exit to use.

In our project Public→IGW; app→same-AZ NAT; DB→local only.

NAT Gateway

FULL FORM
Network Address Translation Gateway

What it is Allows private resources to start outbound Internet connections without becoming publicly reachable.

Why we used it ECS tasks in private subnets can download dependencies or contact AWS/public endpoints safely.

Easy way to remember Like a one-way exit gate: staff can go out, but strangers cannot directly walk in.

In our project Two NAT Gateways, one per AZ.

Elastic IP

FULL FORM
Elastic IP Address

What it is A static public IPv4 address allocated by AWS.

Why we used it Each NAT Gateway needs a stable public address.

Easy way to remember Like a fixed street number for the outbound gateway.

In our project Two EIPs, one for each NAT Gateway.

Security Group

FULL FORM
Security Group

What it is A stateful virtual firewall attached to AWS resources.

Why we used it It enforces least-privilege network access between CloudFront, ALB, ECS and RDS.

Easy way to remember Like a security guard with a list of exactly who may enter and on which door.

In our project ALB origin only; ECS port 3000 from ALB; DB 3306 from ECS.

5. Containers, Load Balancing & Scaling

Amazon ECR

FULL FORM
Elastic Container Registry

What it is A private AWS registry that stores Docker container images.

Why we used it GitHub Actions needs a secure place to push versioned LangNation images before ECS deploys them.

Easy way to remember Like a warehouse containing approved application packages.

In our project Application repository plus dedicated restore repository.

Amazon ECS

FULL FORM
Elastic Container Service

What it is AWS service for running and managing containers.

Why we used it It manages our Node.js containers without us manually operating individual servers.

Easy way to remember Like a manager that assigns work to container workers and keeps them running.

In our project Cluster runs private Fargate tasks.

AWS Fargate

FULL FORM
AWS Fargate

What it is Serverless compute for containers. AWS manages the underlying servers.

Why we used it We want to run ECS containers without provisioning or patching EC2 instances.

Easy way to remember Like renting ready-to-use workstations without owning or maintaining the hardware.

In our project Application baseline: two private tasks; also used for one-off DB restore.

Task Definition

FULL FORM
ECS Task Definition

What it is A versioned blueprint describing the container image, ports, resources, logs and environment variables.

Why we used it Every deployment creates a reproducible definition of exactly how the app should run.

Easy way to remember Like a recipe card for starting one application worker.

In our project Includes DB/RAG environment and secret references.

ECS Service

FULL FORM
Elastic Container Service Service

What it is Maintains the desired number of ECS tasks and replaces failed tasks.

Why we used it It provides self-healing and stable application capacity.

Easy way to remember Like a supervisor who always makes sure enough workers are on duty.

In our project Desired baseline 2, distributed across two AZs.

ALB

FULL FORM
Application Load Balancer

What it is Layer-7 load balancer that receives HTTP requests and distributes them to healthy targets.

Why we used it It gives ECS a stable entry point and balances traffic between tasks in multiple AZs.

Easy way to remember Like a receptionist who sends each visitor to an available employee.

In our project Internet-facing ALB; direct user bypass blocked by SG.

Target Group

FULL FORM
Application Load Balancer Target Group

What it is A group of backend targets that the ALB can send requests to.

Why we used it It connects the ALB to ECS task IPs and continuously checks their health.

Easy way to remember Like the receptionist's live list of employees who are currently available.

In our project Target type ip, port 3000, /health check.

Application Auto Scaling

FULL FORM
Application Auto Scaling for ECS

What it is Automatically changes the number of ECS tasks based on load.

Why we used it It allows the application to handle higher traffic while keeping a controlled minimum and maximum.

Easy way to remember Like calling extra staff when the queue gets busy and sending them home when demand falls.

In our project Min 2, max 4, target CPU 50%.

6. Database, Secrets & Object Storage

Amazon RDS for MariaDB

FULL FORM
Relational Database Service

What it is A managed AWS service for relational databases such as MariaDB, MySQL and PostgreSQL.

Why we used it We need a reliable private database without managing database servers ourselves.

Easy way to remember Like renting a professionally managed records system instead of building the database server yourself.

In our project MariaDB 11.8.9, db.t3.micro, Multi-AZ, 20 GiB gp3, 7-day backups.

Multi-AZ

FULL FORM
Multiple Availability Zones

What it is RDS keeps a synchronous standby database in another Availability Zone.

Why we used it It improves database availability if the primary AZ or database instance has a problem.

Easy way to remember Like keeping a synchronized backup office in another building ready to take over.

In our project Final database architecture remains Multi-AZ.

Secrets Manager

FULL FORM
AWS Secrets Manager

What it is Securely stores sensitive credentials and secrets.

Why we used it Database passwords should not be hard-coded in Terraform or application code.

Easy way to remember Like a locked vault that only authorized services can open.

In our project RDS manages the master password; restore/ECS retrieve references.

Amazon S3

FULL FORM
Simple Storage Service

What it is Highly durable object storage for files and application data.

Why we used it It gives LangNation private storage for future uploads and application objects.

Easy way to remember Like a secure cloud filing cabinet for files rather than database rows.

In our project Private/versioned/AES256/HTTPS-only storage.

7. Event-Driven & RAG Services

Amazon EventBridge

FULL FORM
Amazon EventBridge

What it is Matches and routes AWS events.

Why we used it Filters S3 Object Created events into the document pipeline.

Easy way to remember A smart event dispatcher.

In our project Routes intended storage events to SQS.

Amazon SQS

FULL FORM
Simple Queue Service

What it is Durable message queue.

Why we used it Decouples S3/EventBridge from Lambda and supports retry timing.

Easy way to remember A waiting line that does not lose customers.

In our project Main queue: 4-day retention, 180 s visibility, SSE, maxReceiveCount 3.

Dead Letter Queue

FULL FORM
Dead Letter Queue

What it is Stores repeatedly failing messages.

Why we used it Prevents endless retries and preserves failures for investigation.

Easy way to remember A problem tray for jobs needing manual attention.

In our project 14-day retention; validated with controlled fail/ objects.

AWS Lambda

FULL FORM
AWS Lambda

What it is Serverless function compute.

Why we used it Processes document events without a permanent worker service.

Easy way to remember A worker called only when work arrives.

In our project Python 3.12 processor reads/chunks/embeds knowledge documents.

Amazon Bedrock

FULL FORM
Amazon Bedrock

What it is Managed access to foundation models.

Why we used it Provides embeddings and grounded answer generation without hosting models.

Easy way to remember A managed AI model gateway.

In our project Titan embeddings + Claude Haiku 4.5.

Titan Text Embeddings V2

FULL FORM
Amazon Titan Text Embeddings V2

What it is Converts text to numeric semantic vectors.

Why we used it Makes document chunks and questions comparable.

Easy way to remember Coordinates that describe meaning.

In our project Model amazon.titan-embed-text-v2:0; 1024 normalized dimensions.

Amazon S3 Vectors

FULL FORM
Amazon S3 Vectors

What it is Stores/query vectors using similarity search.

Why we used it Provides the vector knowledge layer for Lina.

Easy way to remember A semantic index rather than an alphabetic index.

In our project langnation-knowledge index, float32, 1024 dims, cosine distance.

8. DNS, HTTPS, Edge Security & Authentication

DNS

FULL FORM
Domain Name System

What it is The Internet naming system that translates human-friendly names into service destinations.

Why we used it Users should access a stable domain instead of remembering changing AWS addresses.

Easy way to remember Like the Internet's phone book.

In our project langnation.com/de is the public cloud endpoint.

Route 53

FULL FORM
Amazon Route 53

What it is AWS authoritative DNS service.

Why we used it We use it for reliable DNS and to point langnation.com/de to the current CloudFront distribution.

Easy way to remember Like the official address directory for the domain.

In our project Personal AWS hosted zone points langnation.com/de to current CloudFront.

TLS / HTTPS

FULL FORM
Transport Layer Security / Hypertext Transfer Protocol Secure

What it is Encrypts traffic between the user's browser and the application front door.

Why we used it Credentials and API traffic must not travel as plain text.

Easy way to remember Like putting messages inside a locked envelope while they travel.

In our project CloudFront redirects HTTP to HTTPS and uses the ACM certificate. Viewer HTTPS ends at CloudFront; CloudFront reaches the ALB over HTTP port 80.

ACM

FULL FORM
AWS Certificate Manager

What it is Creates and manages TLS certificates used for HTTPS.

Why we used it CloudFront needs a trusted certificate for langnation.com/de.

Easy way to remember Like the verified digital certificate that creates the browser padlock.

In our project Certificate covers langnation.com/de and wildcard.

CloudFront

FULL FORM
Amazon CloudFront

What it is AWS global Content Delivery Network and edge front door.

Why we used it It provides HTTPS, a stable edge endpoint and the layer where WAF protects the application.

Easy way to remember Like a global reception desk close to users that forwards approved requests to the origin.

In our project Custom alias langnation.com/de; origin is ALB over HTTP :80.

AWS WAF

FULL FORM
Web Application Firewall

What it is Inspects HTTP requests and blocks patterns associated with web attacks or abusive traffic.

Why we used it It adds an application-security layer before requests reach the ALB and ECS.

Easy way to remember Like a security scanner at the entrance that checks every visitor's request.

In our project Managed common rule set + rate rule; controlled suspicious test returned 403.

Amazon Cognito

FULL FORM
Amazon Cognito

What it is Managed user authentication service for sign-up, sign-in and token issuance.

Why we used it We need a secure login system without building password storage and token issuance ourselves.

Easy way to remember Like the identity desk that checks a user's login and issues a digital access pass.

In our project Email-based User Pool; /protected requires access token.

JWT

FULL FORM
JSON Web Token

What it is A signed token containing claims that can prove a user has authenticated.

Why we used it The Express API can validate the Cognito access token before allowing protected requests.

Easy way to remember Like a signed temporary digital ID card.

In our project aws-jwt-verify validates Cognito access tokens.

9. Monitoring, Governance, Release Security & Cost

Amazon CloudWatch

FULL FORM
Amazon CloudWatch

What it is AWS monitoring service for logs, metrics and operational visibility.

Why we used it We need logs from ECS and monitoring data to troubleshoot and prove the system is healthy.

Easy way to remember Like CCTV, dashboards and a logbook for the cloud platform.

In our project Operations dashboard plus nine custom alarms.

AWS CloudTrail

FULL FORM
AWS CloudTrail

What it is Records AWS management API activity.

Why we used it Provides durable audit evidence.

Easy way to remember An audit log of administrative actions.

In our project Multi-Region trail → private versioned encrypted S3 bucket.

Trivy

FULL FORM
Trivy

What it is Scans container images for known vulnerabilities.

Why we used it Blocks CRITICAL image vulnerabilities before ECS deployment.

Easy way to remember A security inspection before shipping.

In our project Detected real CRITICAL tar vulnerability; release fixed by upgrading to Node 24.

AWS Cost Explorer

FULL FORM
AWS Cost Explorer

What it is Analyzes AWS cost and usage.

Why we used it Shows where architecture spend actually comes from.

Easy way to remember The cloud bill analyzer.

In our project October usage subtotal ≈ $23.78; NAT hourly cost was an important driver.

10. Important Design Decisions to Remember

Decision	Reason
Why persistent Route 53?	The training sandbox is disposable, but the personal AWS hosted zone persists. Only the cloud subdomain target needs to follow the recreated CloudFront distribution.
Why ECS and RDS are private?	They should not accept direct internet traffic. Users traverse CloudFront/WAF/ALB; MariaDB accepts 3306 only from ECS.
Why direct ALB access is blocked?	Otherwise users could bypass CloudFront and WAF. ALB ingress is restricted to the CloudFront origin-facing prefix list.
Why GitHub OIDC instead of AWS keys?	GitHub receives short-lived STS credentials only when a workflow runs; no long-lived deployment key is stored.
Why HCP Terraform?	The local state is not authoritative. HCP stores persistent remote state and is the trusted plan/apply/rebuild path.
Why Git-SHA image tags?	Running source commit, ECR image and ECS task revision can be traced exactly.
Why SQS between events and Lambda?	Buffering decouples producer/consumer and gives controlled retry/DLQ behavior.
Why 200-word chunks with 40-word overlap?	Simple explainable balance between context continuity and retrieval granularity.
Why the same Titan embedding model for ingestion and query?	Document and question vectors must exist in the same semantic space.
Why verified RDS TLS?	RDS secure transport is a security requirement; the application should adapt rather than weakening the DB.
Why quality gates before deployment?	Tests, Terraform validation, smoke test and vulnerability scanning should block release before ECS changes.
Why keep two NAT Gateways and Multi-AZ RDS after cost review?	They support the production-style availability story; lower-cost development options can be modeled separately.

11. Important Numbers and Final Values

Item	Value
AWS Region	us-east-1
VPC	10.0.0.0/16
Public subnets	10.0.1.0/24, 10.0.2.0/24
Private app subnets	10.0.11.0/24, 10.0.12.0/24
Private DB subnets	10.0.21.0/24, 10.0.22.0/24
ALB → ECS	TCP 3000
ECS → MariaDB	TCP 3306
ECS Auto Scaling	min 2, max 4, target CPU 50%
MariaDB	11.8.9, db.t3.micro, Multi-AZ
SQS visibility timeout	180 seconds

Item	Value
SQS maxReceiveCount	3
DLQ retention	14 days
Embedding model	amazon.titan-embed-text-v2:0
Embedding dimensions	1024
Chunking	200 words / 40-word overlap
Public domain	langnation.com/de
M12 recovery observation	~44 seconds from 2/1/1 back to 2/2/0
October usage subtotal	~$23.78 before tax
October shown total	~$28.07 including $4.29 tax

12. Flows You Should Be Able to Explain

A. User request flow

```
User → Route 53 → CloudFront/ACM → WAF → ALB → Target Group
  → private ECS Fargate → real LangNation app.
```

B. Deployment flow

```
Developer → git push → quality gates → GitHub OIDC/STS → Docker build → ECR
  → ECS revision → rolling deploy → final verification.
```

C. Knowledge ingestion flow

```
S3 upload → EventBridge → SQS → Lambda → S3 GetObject → chunk → Titan embedding
  → S3 Vectors.
```

D. Lina RAG flow

```
Question → Titan query embedding → S3 Vectors search → source_text
  → Claude Haiku → grounded answer.
```

E. Monitoring/audit flow

```
Runtime metrics/logs → CloudWatch dashboard/alarms; AWS management actions
  → CloudTrail → private audit S3.
```

F. Morning recovery flow

```
Manual sandbox activation → HCP rebuild → DB restore/58-table validation
  → app deploy → CloudFront output → Route 53 UPSERT/INSYNC → public /health.
```

G. Resilience flow

```
2/2/0 baseline → stop one ECS task → 2/1/1 while HTTP 200
  → automatic replacement → ~44 s → 2/2/0.
```

13. Very Short Memory Trick

Memory word	Services / concept
NETWORK	VPC, subnets, routes, NAT, security groups
CONTAINER	Docker → ECR → ECS Fargate
TRAFFIC	Route 53 → CloudFront/WAF → ALB → target group
DATA	MariaDB + S3 + Secrets Manager
EVENTS	S3 → EventBridge → SQS → Lambda → DLQ
AI/RAG	Titan → S3 Vectors → Claude Haiku
IDENTITY	IAM/OIDC for automation; Cognito/JWT for users
AUTOMATION	Terraform/HCP + GitHub Actions
OBSERVABILITY	CloudWatch dashboard + alarms
GOVERNANCE	CloudTrail → audit S3
QUALITY	Tests + Terraform validation + smoke test + Trivy
RESILIENCE/COST	ECS self-healing + Cost Explorer

14. One-Minute Explanation

LangNation runs inside a private AWS VPC across two Availability Zones. Users enter through persistent Route 53 and CloudFront over HTTPS, with ACM providing the certificate and AWS WAF filtering web traffic. CloudFront is the only allowed origin source to the Application Load Balancer, which distributes requests to private ECS Fargate tasks. The real application uses a private Multi-AZ MariaDB database over verified TLS, private S3 storage and Cognito authentication. The RAG ingestion pipeline uses S3, EventBridge, SQS and Lambda; Titan creates embeddings, S3 Vectors performs semantic retrieval, and Claude Haiku generates Lina's grounded answers. Terraform and HCP Terraform manage infrastructure, while GitHub Actions uses OIDC, application/Terraform/container quality gates and immutable Git-SHA images for deployment. CloudWatch provides operational monitoring, CloudTrail provides audit history, and OP01 can restore the infrastructure, sanitized 58-table database, application and Route 53 endpoint after the training sandbox is manually activated. The final resilience test proved the public endpoint remained healthy while ECS automatically replaced a stopped task.

> **Best way to study AWS:** Do not memorize definitions word-for-word. For each service remember: (1) what problem it solves, (2) where it sits in our architecture, and (3) what would break if we removed it.

— End of guide —